Crop Science
Lecture 5
30th August 2021

Crop productivity

Canopy architecture can be viewed as a visible but momentary expression of plant form

architecture refers to the spatial configuration of the aboveground plant organs

plant canopy architecture is, in many ways, an abstraction that relates plant form to an orderly, genetically determined “growth plan

Unlike buildings, plant canopies are dynamic and constantly changing with time and space in response to their environment

Plant canopy

A primary function of plant canopies is to intercept radiation to drive photosynthesis and other metabolic processes

This interception occurs with varying degrees of efficiency, determined chiefly by leaf area, configuration relative to the sun, and to a lesser extent the spatial arrangement of leaves

Other important morphological canopy characteristics include plant height, branching patterns, and sizes and shapes of individual leaves

At least 90% of the biomass of higher plants is derived from CO2 assimilated through photosynthesis

canopy that lead to greater interception of sunlight increase photosynthesis, and grain or dry matter yield. Conversely, those that lead to poor light interception result in lower growth and yield

When speaking of plant canopies or other plant features, it is useful to have a mental picture of an ideal form

This process continues today with modern plant breeders, who often have a mental concept of an ideal plant, or ideotype, that guides their breeding programs

Ideotype is a biological model that is expected to perform or behave in a predictable manner within a defined environment

crop ideotype” is a plant model that is expected to yield a greater quantity and/or quality of grain

It should be apparent that what is ideal for one environment would not be ideal for another. For example, a plant for hot and dry environments may not be ideal for cold and wet environments

Plant height

Plant height is the distance from ground level to the tip of the tallest leaf for seedlings or juvenile plants

Mature plants, it is the distance from ground level to the tip of the tallest panicle, ear, or head in cereals and branch in legumes

Short and sturdy shoot, more than any other character, favor lodging resistance

Lodging is the permanent displacement of the stems from their upright position

There are three types of lodging: breaking of the stem, bending of the stem, and rolling,

Plant lodging

Further, strong winds and rains during reproductive and grain tilling stages of growth can cause lodging in annual crops. Lodging during the grain filling growth stage can reduce grain quality

A short and sturdy shoots also promotes favorable grain-to-straw ratios, adequate N responses, and high yield capacities

The creation of semi-dwarf cultivars spectacularly increased the yielding ability of many crops, such as rice and wheat

Lodging resistance is related to short stature, but also dependent on other characters like shoot diameter, shoot wall thickness, and the degree to which leaf sheaths wrap internodes

Therefore, an optimum plant height is important, which may vary with plant species and environment

Plant branches

Branches develop from the main shoot or from other shoots during vegetative growth

The development of buds after differentiation is greatly affected by environmental conditions, as well as by genotypic characteristics

It is essential that sufficient supplies of water, photosynthate, nutrients, and plant hormones are present, and that stress is minimal

Once leaves develop from the branch, they can perform photosynthesis and produce carbohydrates

For many plant species, tillering is a mechanism for assuring that the amount of biomass produced is in balance with the amount of plant resources available

Leaf Characteristics

Leaf characters such as erectness, length, width, thickness, toughness, color, and senescence are often associated with yielding ability of plant species or cultivars

Among these characters, erect orientation of leaves is associated with high yielding capacity

Erect leaves permit greater penetration and more even distribution of light into the crop, and thus higher photosynthetic activity

Plants with upright leaves may require less physical space than plants with horizontal leaves and may be adapted to narrow row widths

Thickness

Leaf thickness is also a desirable leaf trait

Thicker leaves have higher density of chlorophyll per unit leaf area, and have greater photosynthetic capacities than thinner leaves

Vigorous leaf growth in crop plants has generally been associated with long-term gains in photosynthetic potential

Plant size may provide substantial yield benefits. For example, rice leaf growth rate is lower compared with maize

Leaf size is directly associated with leaf angle, with short leaves tending to be more erect than longer ones

As leaves become longer, they become more difficult for the midrib to support, and therefore tend to droop. Usually, dwarf cultivars have short leaves and tall cultivars have long leaves

These strong associations indicate that leaf length in both dwarf and tall cultivars is an effect of genes for plant height

Examples of higher yield being associated with more erect leaf angle distribution

Similarly, maize yield has been greater for cultivars with erect canopies than for cultivars with more horizontal orientation

Light entering a canopy of erect leaves is spread over larger photosynthetic areas than in prostrate cultivars, resulting in greater photosynthetic efficiency

Leaf width is less variable than length

Narrow leaves are more desirable than the wider ones due to more even distribution in the plant canopy

Thick leaves are associated with high yielding capacities of crop cultivars

Leaf thickness can be measured microscopically, but it is conveniently expressed as specific leaf area (area per mass) or specific leaf weight (mass per area)

Other leaf characters that have been associated with yield include leaf toughness, color, and senescence.

Leaf toughness, which is important in preventing breaking during heavy wind and rain, is related to thickness and lignification of leaf tissues

Senescence refers to degenerative changes normally associated with the declining phase of leaf, whole plant physiological activities, or both

Senescence is of economic interest because it affects crop productivity by reducing the active photosynthetic area

Summary

Aboveground plant architecture and its components, including plant height, tillering or branching, leaf size, shape and arrangement, are important plant morphological characters related to yield

The main physiological function of the canopy is interception of light and subsequently plant photosynthetic activity

Distribution of leaf area within a plant canopy is a major factor determining total light interception, which affects photosynthesis, transpiration, and dry matter accumulation

Vertical distribution of leaf area is determined by leaf size, leaf angle, and internodes length

Plant height is also related to plant lodging resistance, a character that determines plant yield capacity and grain quality

Some desirable plant architecture characters related to high yield are semi-dwarf plant height, thick, short, small and erect leaves, short and stiff shoots, upright (compacted) leaves, high N response, and high grain-to-straw ratio